



Effluent to Essence

Repurposing Acidic Effluent for the Future of Polymer Recycling

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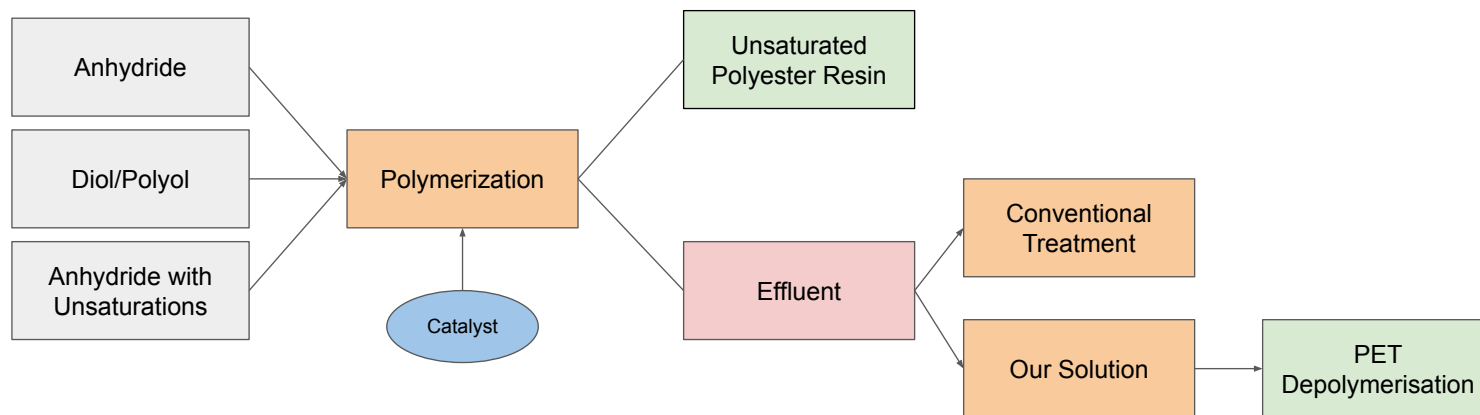


Problem Statement

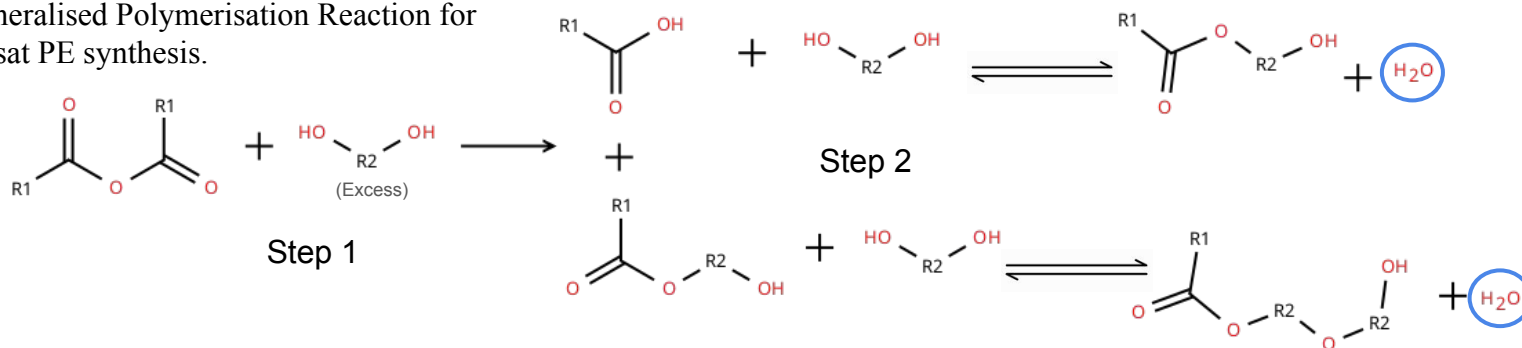
The company is an unsaturated polyester resin manufacturer. During their production process, acids and glycols are reacted to synthesize unsaturated polyester resins, and as a result, water is generated as a by-product. This water forms part of the process effluent and is characterized by very high Biological Oxygen Demand (BOD) and Chemical Oxygen Demand (COD), a strong acidic odor, and a highly acidic pH of around 2. The company is interested in exploring innovative ways to utilize this effluent instead of traditional disposal or treatment methods. Students are invited to propose practical and technically feasible solutions to reuse or convert this effluent into value-added applications within the polymer and coatings industry, while considering industrial feasibility, safety, and sustainability.



Flowchart of the Problem Statement



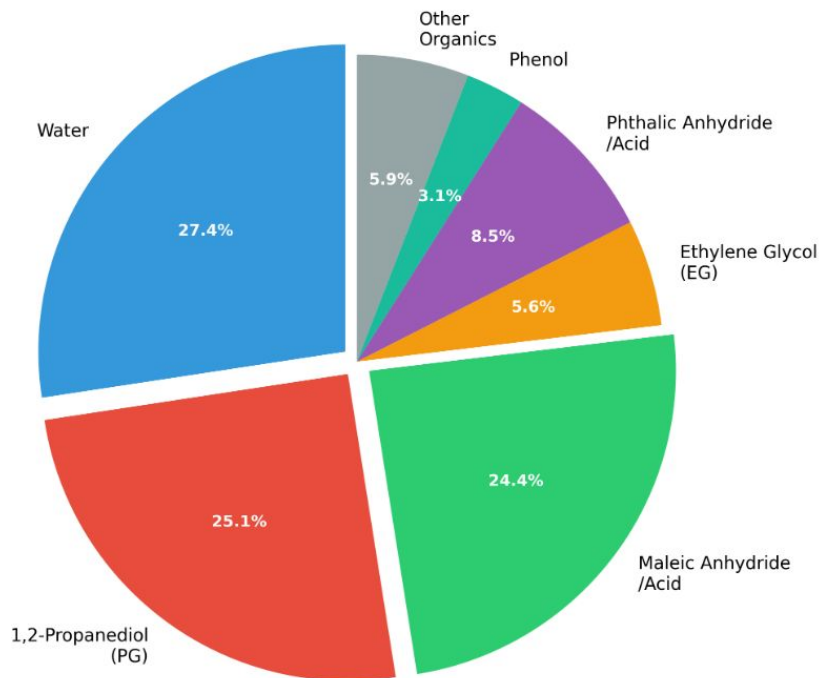
Generalised Polymerisation Reaction for Unsat PE synthesis.





Composition & Preliminary Steps

**Typical UPR Production Effluent Composition
(High Acidity, pH ~2)**

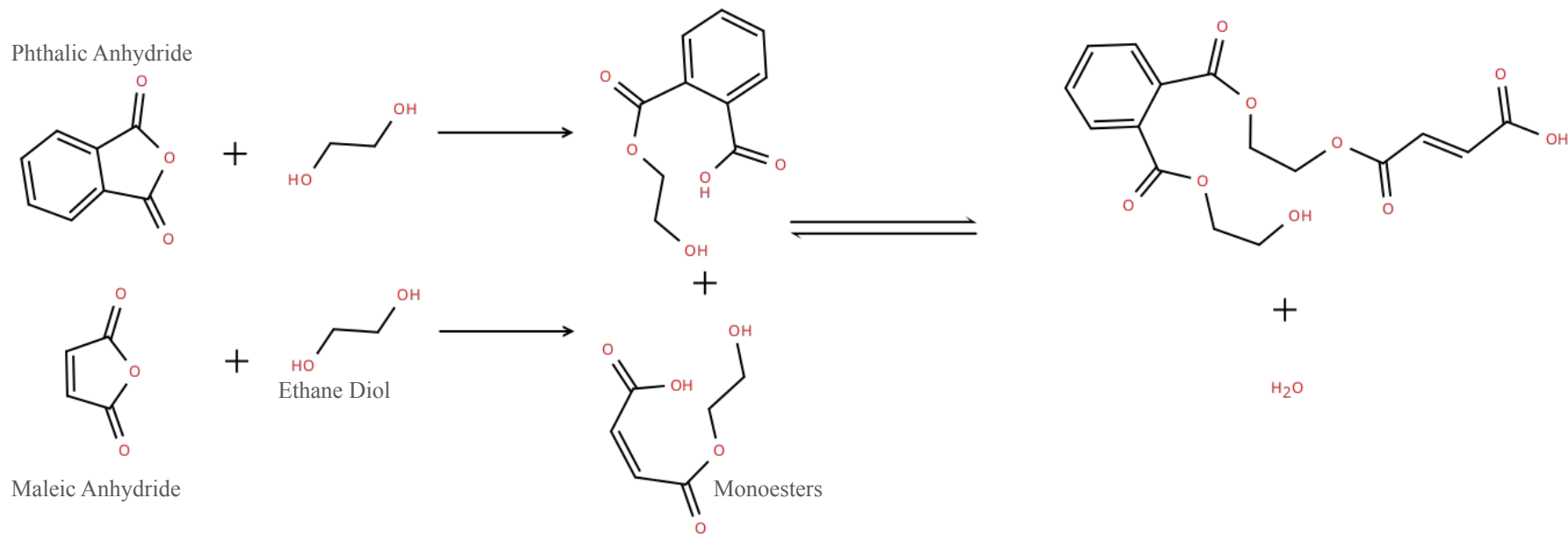


To tackle this problem statement, we first need to understand more about this effluent and its composition, such as what molecules are increasing the BOD/COD?, which molecule is giving off the strong Acidic smell? and so on. This will require characterization techniques such as Spectroscopy, Chromatography and other known characteristics techniques. We ensure our sample does not damage the Characterization equipment by following Standard Operating Procedures.

One of many cases where Unsaturated Polyester is Synthesised by the following

1. Phthalic Anhydride
2. Maleic Anhydride
3. Ethylene 1,3 Diol
4. Uncatalysed reaction (ref: <https://doi.org/10.1002/pen.760331809>)

Reaction representation based this case-



Acidolysis of PET catalysed by TPA

2.5 g of PET, a TPA concentration of 0.1 g/mL, mass ratio PET:H₂O of 1:8, 220 °C of temperature, and 180 min of reaction time, a PET conversion of up to 100.0% and a TPA yield of 95.5% were achieved.

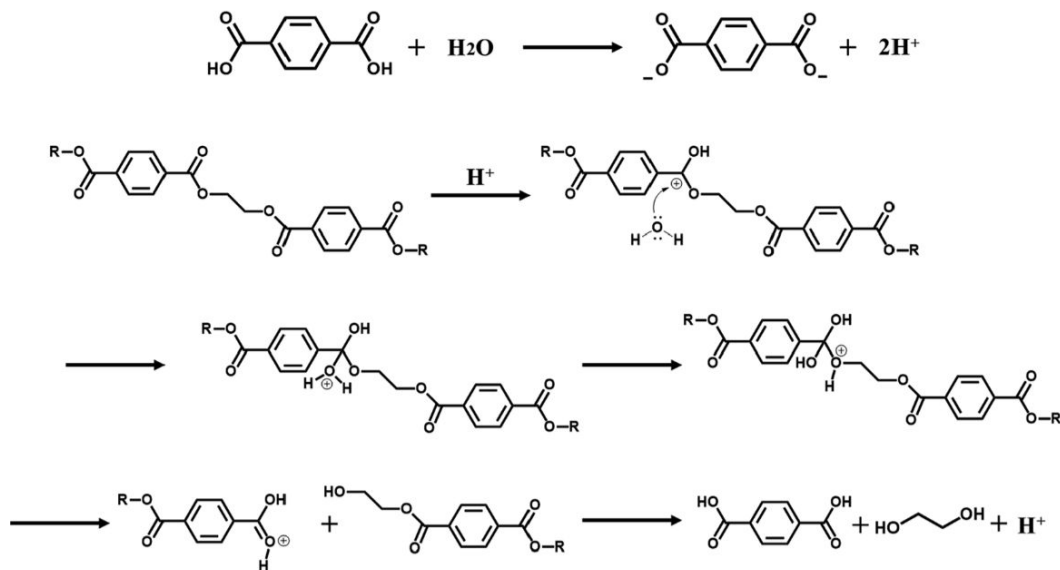
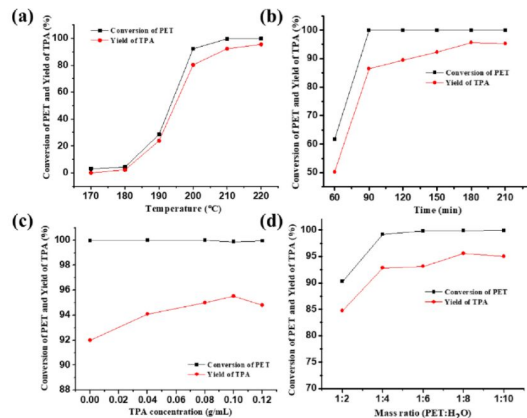


Fig. 2. PET hydrolysis mechanism with TPA as an acid catalyst



Adopted from
<https://doi.org/10.1016/j.wasman.2021.09.009>

Ref for H₂SO₄ catalysed:
<https://doi.org/10.1177/14777606221128038>

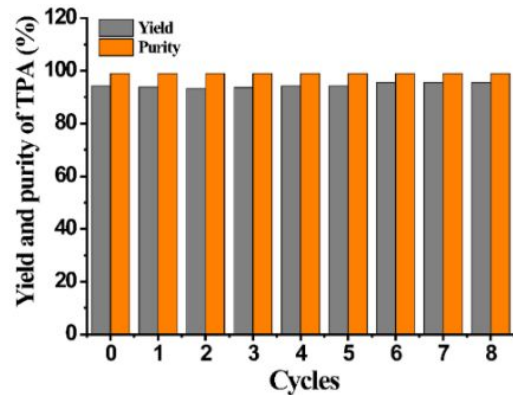
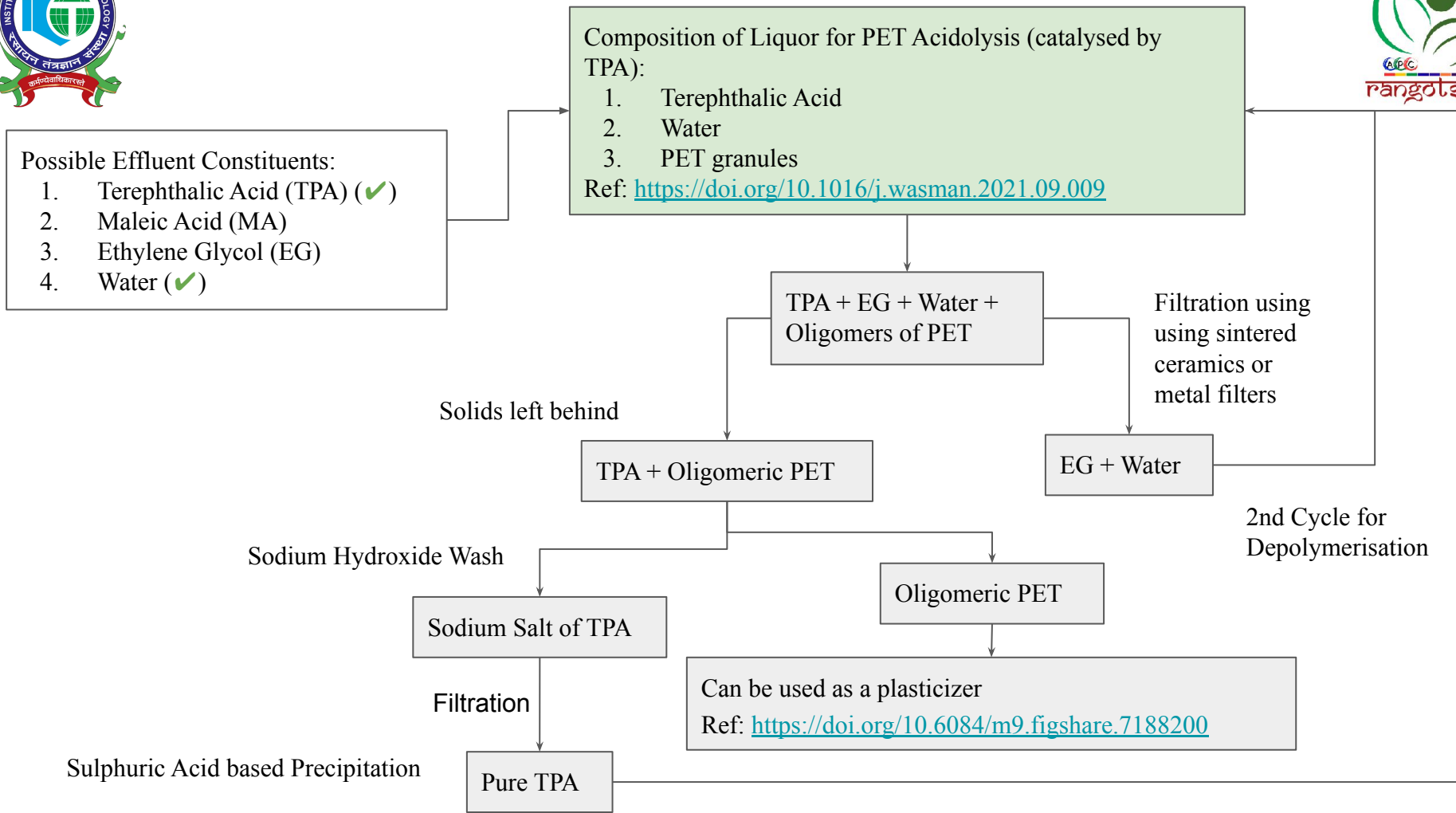


Fig. 7. Recycling and reuse of terephthalic acid (TPA) for polyethylene terephthalate (PET) hydrolysis (reaction conditions: 2.5 g of PET, TPA concentration of 0.1 g/mL, PET:H₂O mass ratio of 1:8, 220 °C, 180 min).





Thank You for your Attention
Questions are Welcomed